INCH-POUND

MIL-S-48687 (AR)
AMENDMENT 2
16 March 1994
SUPERSEDING
AMENDMENT 1
16 February 1990

MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143E1
LOADING, ASSEMBLING AND PACKING FOR

This amendment forms a part of Military Specification MIL-S-48687 (AR), dated 4 February 1987, and is approved for use by the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 2.1.1, Under STANDARDS, Delete: "MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for" and substitute "MIL-STD-2000 - Soldered Electrical and Electronic Assemblies, Standard Requirements for".

PAGE 7

Table IV, g & h: Delete "2.6" (six places) and substitute "4.0" (six places).

PAGE 8

3.7.1, Line 7: Delete "2.5" and substitute "1.5".

3.7.2, Last line: Delete "2.5" and substitute "4.0".

3.7.4, Delete in its entirety and substitute the following:

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"3.7.4 Fire circuit function. The initiation of either the left or right fire circuits functioning is determined by the occurrence of the fire TM output changing from 0 ± .5 VDC to 4.5 ± 1.0 VDC at (+) J1-20 to (-) J1-15. The left fire circuit has completed functioning when the left fire circuit capacitor voltage (+) J1-38 to (-) J1-15 has discharge to less than 15.0 VDC in no greater than 400 milliseconds after the occurrence of a fire TM output. The right fire circuit has completed functioning when the right fire circuit capacitor voltage (+) J1-36 to (-) J1-15 has discharge to less than 15.0 VDC in no greater than 400 milliseconds after the occurrence of a fire TM output."

PAGE 9

3.7.4.1.1, Delete in its entirety and substitute the following:

"3.7.4.1.1 Loss of power after arming. The initiation of fire circuit functioning as defined by paragraph 3.7.4 shall occur in no less than 10 milliseconds and no greater than 50 milliseconds upon removal of power/ destruct (+) J1-30 to (-) J1-15."

3.7.4.1.2, Delete the last sentence and substitute the following:

"The fire circuits shall function as described in paragraph 3.7.4. Both the left and right fire circuit capacitor voltages shall have discharge to less than 15.0 VDC in 1.2 seconds maximum after the initial fire TM voltage transitioned as described above."

* 3.7.4.2.1, Line 2: Delete "5.0 ± 0.25V" and substitute "4.50 ± 0.25V".

3.7.4.2.4, Line 4: Delete "then".

PAGE 12

3.9.1, Table VI, under minimum and maximum.

(OHMS) Delete:	"6.1	8.2	and substitute	"5.9	8.1
	4.8	6.5		4.6	6.2
	7.2	9.2"		7.3	9.1"

* 3.9.3, Last line: Delete: "± 300 milliseconds" and substitute "within ± 500 milliseconds".

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3.12.4, Delete in its entirety and substitute the following:

"3.12.4 Lateral arming. When a Safe and Unlocked S&A Device with 5.0 + 1.0 VDC applied to 5V Test (+) J1-1 to (-) J1-15, is subjected to the acceleration environment specified below, both detonator rotors shall arm in 2.8-4.0 seconds and remain Armed for the duration of the application of the acceleration. Arming is defined as the detonator rotors rotating from the Safe and Unlocked position to the Armed position when Rotor Monitor (Right) (+) J1-18 to (-) J1-15 and Rotor Monitor (Left) (+) J1-2 to (-) J1-15 have changed from 4.5 ± 1.0 VDC to 0.0 ± 0.5 VDC. A minimum of one Detonator Rotor shall be Armed and Locked after the removal of the acceleration. The side required to be locked shall be the (left) side when the direction of the applied lateral acceleration is in the (+Y) axis. The side required to be locked shall be the (right) side when the direction of the applied lateral acceleration is in the (-Y) axis (refer to Figure 1). The side required to be locked shall be locked in ± 500 milliseconds of the arming time of that detonator rotor.

a. An acceleration profile along the first transverse axis (+Y, -Y) of the S&A which is equal to the lateral acceleration profile in drawing 9357707.

b. An acceleration along the longitudinal axis of the S&A in the (-Z) direction, which is equal to the axial acceleration profile in drawing 9357707.

c. Acceleration components to be applied simultaneously to the S&A. Maximum permissible variation in centrifuge revolutions per minute (RPM) is ± 8%. Maximum permissible variation in angular orientation of the S&A is ± 4% (See 4.5.11.1.1)."

3.12.5, Delete: "a and b" and substitute

"Vector sum of:

a. 46.8g longitudinal aft-to-forward (-Z) and 52.0g lateral (+Y or -Y)
(=70.0g at 48 from longitudinal axis).

b. 22.3g longitudinal forward-to-aft (+Z) and 52.5g lateral (+Y or -Y)
(=57.0g at 67 from longitudinal axis)."

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3.14.11, Last line: Delete "2.5" and substitute "1.5".

PAGE 22

3.14.14: Delete in its entirety.

PAGE 25

4.3.1, a, Under part description: Delete "Electrical Assembly" and substitute "Detonator Electrical Assembly".

4.3.1, m: Delete in its entirety and substitute "EM assembly, 12630801, 10".

4.3.1. o, Under part description: Delete "M143" and substitute "M143E1".

PAGE 27

4.3.4, Under paragraph reference/inspection method: Delete "4.4.2.31" and substitute "4.4.2.33".

PAGE 29

4.4.2.1, Sheet 1 of 1, next higher assembly: Delete "12630800" and substitute "12630858".

PAGE 30

4.4.2.2, Sheet 1 of 1, Major 102, under Examination or Test: Add "(2 places)".

PAGE 31

* 4.4.2.3, Sheet 1 of 1, Major 103, under Examination or Test, Delete: "MIL-STD-1460" and substitute "MIL-STD-2000".

PAGE 38

4.4.2.10, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

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* 4.4.2.12, Sheet 1 of 1, Major 101, under Examination or Test, Delete: "MIL-STD-1460" and substitute "MIL-STD-2000".

PAGE 42

4.4.2.14, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 43

4.4.2.15, Sheet 1 of 1: Delete in its entirety.

PAGE 44

4.4.2.16, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 46

4.4.2.18, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 48

4.4.2.20, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

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QUALITY CONFORMANCE INSPECTION

MIL-S-48687 (AR)
AMENDMENT 2

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Lead Assembly		9311077
			NEXT HIGHER ASSEMBLY 12630861

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Evidence of cracks at crimp		100%	3.2	Visual
Major					
101	Length		0.40%	3.2	Gage
102	Diameter, max.		0.40%	3.2	Gage
103	Radiographic Inspection		0.40%	3.5.1	MIL-STD-202, M209
104	Output		0.40%	3.5.2	4.5.10.2/Test
Minor					
201	Missing or excessive green lacquer		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.14	Electro-Mechanical Assembly (Visual Prior to Installing Cover Assembly)		12630801
			NEXT HIGHER ASSEMBLY 12630864/12630800

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Fire board mounting screws missing or loose (8 places)		0.40%	3.2	Visual/Manual
102	Soldering not in accordance with MIL-STD-2000		0.40%	3.2	Visual/MIL-STD-2000
Minor					
201	Board stiffener washer missing (1 place)		0.65%	3.2	Visual
202	Board stiffener insulated washers missing (3 places)		0.65%	3.2	Visual
203	Conformal coating loose or flaking		0.65%	3.18	Visual
204	Evidence of poor workmanship		1.0%	3.18	Visual

NOTES:

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MIL-S-48687 (AR)
AMENDMENT 2

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Gasket				12630868
					NEXT HIGHER ASSEMBLY
					12630801
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter of hole (4 places)		0.65%	3.0	Gage
202	Location of hole (4 places)		0.65%	3.0	Use P/N 12630849
203	Dimensions of cut-out		0.65%	3.0	as a functional
204	Location of cut-out		0.65%	3.0	gage 1/
205	Evidence of poor workmanship		1.0%	3.18	Visual

NOTES:

1/ 50% or more of each of the 4 mounting holes shall be visible through the gasket.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.18	Safety & Arming Device: M143E1 (Loaded) After Installing Lead Assembly				12630800
					NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lead assembly mounting screws missing or loose		0.40%	3.2	Visual/Manual
102	Logic board mounting screws missing or loose (4 places)		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48687 (AR)
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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Electro-Mechanical Assembly (Visual/Test Prior to Electrical/Mechanical Operation)				12630801
					NEXT HIGHER ASSEMBLY 12630864/12630800
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mechanical shorting plate detonator housing assembly not replaced with resistor shorting plate detonator housing assembly (2 places)		100%	3.6.1	Visual
102	Resistance (2 places)		100%	3.6.1	Test

NOTES:

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4.4.2.21, Sheet 1 of 1, Major 105, under requirement paragraph: Delete "3.9.1" and substitute "3.9.1a".

PAGE 53

4.4.2.25, Sheet 1 of 2, Major 106, under requirement paragraph: Delete "3.9.3".

PAGE 54

4.4.2.25, Sheet 2 of 2, Under Examination or Test: Delete "12630800" and substitute "12630865-3".

PAGE 56

4.4.2.27, Shee 1 of 1, Major 105 and 106: Delete in their entirety.

PAGE 57

4.4.2.28, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 60a

Add: "4.4.2.32, Sheet 1 of 1 with included sheet".

PAGE 60b

Add: "4.4.2.33, Sheet 1 of 1 with included sheet".

PAGE 62

4.5.2, title: Delete in its entirety and substitute the following:

"Procedure for Function on Loss of Power after Arming Test".

4.5.2.1: Add "e. 5.0 + 1.0 VDC Applied to 5V Test input".

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.28	Safety & Arming Device: M143E1 (Loaded) Electrical Operation, CRAF Test Set	1	1	12630800
				NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED				
Critical					
1	Safety (Safe & Locked)	100%			Test/4.5.1.2
Major					
101	Continuity	100%		3.14.1	Test
102	Resistance	100%		3.14.2/ 3.9.1a	
103	Connector Pins to Case Ground Isolation Resistance	100%	1/	3.14.3	Test
104	Fire Circuit Charging (charge command)	100%		3.14.11	Test
105	Fire Circuit Capacitor Voltage Storage	100%		3.14.12	Test
106	Voltage Outputs (Safe & Locked)	100%		3.14.13	Test
107	Fire Circuit Discharging	100%		3.14.14	Test

NOTES:
1/ Accept on 0 defects, reject on 1 or more defects.

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QUALITY CONFORMANCE INSPECTION

MIL-S-48687 (AR)
AMENDMENT 2

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.32	LAT Shorting Plate Detonator Housing Assembly	1 OF 1	9399905-01
			NEXT HIGHER ASSEMBLY 12630865

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Detonator not flush after assembly	1/	100%	3.2	Visual
202	Detonator leads not twisted together		100%	3.2	Visual
203	Insulation sleeving missing		100%	3.2	Visual
204	100 ohm resistor missing or incorrect value		100%	3.2	Visual
205	Evidence of solder flow into contact area		100%	3.2	Visual
206	Evidence of poor workmanship		100%	3.18	Visual

NOTES:
1/ Two times the number of LAT samples.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48687 (AR)
AMENDMENT 2

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.33	LAT Shorting Plate Detonator Housing Assembly	1	1	9399905-02
				NEXT HIGHER ASSEMBLY 12630865

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Detonator not connected	1/	100%	3.2	Visual
202	Insulation sleeving missing		100%	3.2	Visual
203	Solder not in accordance with MIL-STD-2000		100%	3.2	Visual
204	Evidence of solder flow into contact area		100%	3.2	Visual
205	Evidence of poor workmanship		100%	3.2	Visual

NOTES:
1/ Two times the number of LAT samples.

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4.5.2.2: Delete "a. and b." and substitute:

"a. Perform arming as specified in 4.5.7.2.

b. Remove 31.0 + 0.5 VDC on Power/Destruct.

c. Measure Time (3.7.4.1.1) to initiation of Fire Circuit Functioning.

d. Measure Time (3.7.4) for L & R Fire Circuits to Function".

4.5.3.1: Add "f. 5.0 + 1 OVDC Applied to 5V Test input".

4.5.3.2, Under b.: Delete "4.5.7" and substitute "4.5.7.2".

4.5.3.2, Under c.: Delete in its entirety and substitute

"c. Measure Time (3.7.4.1.2) to change of rotor monitor with respect to fire TM and verify capacitor voltage left and right is less than 15.0 VDC at 1.2 seconds after initial occurrence of fire TM."

4.5.3.2: Add "d. Measure Time (3.7.4) for left and right Fire Circuits to Function."

PAGE 64

4.5.8.2, Line 7: Delete "both detonators" and substitute "both detonator rotors".

PAGE 66

4.5.10: Delete "." and substitute "/S&A (loaded)".

4.5.10.1, Line 1, after Assembly: Add "or S&A (loaded)".

4.5.10.1, CAUTION, line 1, after Assembly: Add "or S&A (loaded)".

4.5.10.2, Line 1, after assembly: Add "or S&A (loaded".

4.5.10.2, CAUTION, line 1, after Assembly:

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4.5.11.1.1: Add "The recommended method for demonstrating compliance to these requirements is sampling both R.P.M. and angular orientation in intervals no greater than 50 milliseconds throughout the duration imparted to the S&A device."

PAGE 71

5.2, a, b, & c: Delete in its entirety.

6.2.2, Last line: Delete "Dover, New Jersey 07801-5001" and substitute "Picatinny Arsenal, New Jersey 07806-5000".

PAGE 72

6.6: Delete " $\frac{g \times 10^9}{R \times 28416}$ " and substitute " $187.6 \frac{g}{R}$ ".

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APPENDIX A: Delete in its entirety and replace with included sheet.

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APPENDIX A: Delete in its entirety and replace with included sheet

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APPENDIX A

TABLE I. Traceability and performance

	PART DESCRIPTION	DRAWING	DOCUMENTATION TRACE	DOCUMENTATION PERFORMANCE	DATA REQUIRED MIL-SPEC
a.	S&A Device, Loaded	12630800	REQUIRED	REQUIRED	MIL-S-48687
b.	Detonator	9279861	REQUIRED	REQUIRED	MIL-D-48175
c.	Lead Assembly	9311077	REQUIRED	REQUIRED	MIL-S-48687
d.	Electro-Mechanical Assy	12630801	REQUIRED	REQUIRED	MIL-S-48687
e.	Mechanical Assy	9390590	REQUIRED	REQUIRED	MIL-S-48686
f.	Switch Board	12630822	REQUIRED	NA	NA
g.	Solenoid, Rotary	12630826	REQUIRED	REQUIRED	MIL-S-48686
h.	Switch, Lever	9390596	REQUIRED	NA	NA
i.	Plate Shorting, Det.	12630806	REQUIRED	NA	NA
j.	Plate Shorting, S&A	12630812	REQUIRED	NA	NA
k.	Electronics Assembly	12630829	REQUIRED	REQUIRED	MIL-S-48685
l.	Logic Circuit Board Assy	12630845	REQUIRED	REQUIRED	MIL-S-48687
m.	Capacitor, 68 Microfarad	12630845	REQUIRED	REQUIRED	MIL-S-48685
n.	Logic Circuit Hybrid	9311370	REQUIRED	REQUIRED	MIL-S-63256

NA = Not Applicable

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APPENDIX A

TABLE I. Traceability and performance

	PART DESCRIPTION	DRAWING	DOCUMENTATION TRACE	DOCUMENTATION PERFORMANCE	DATA REQUIRED MIL-SPEC
o.	Fire Board Assy	12630829	REQUIRED	REQUIRED	MIL-S-48685
p.	Capacitor, 86 Microfarad	12630829	REQUIRED	REQUIRED	MIL-S-48685
q.	Diode, 1N4489	12630829	REQUIRED	REQUIRED	MIL-S-48685
r.	Diode, 1N485B JANTX	12630829	REQUIRED	NA	NA
s.	SCR, 2N2326 JANTX	12630829	REQUIRED	NA	NA
t.	Choke	12630831	REQUIRED	REQUIRED	MIL-S-48685
u.	Transformer	12630840	REQUIRED	REQUIRED	MIL-S-48685
v.	Bridge Rectifier	12630860	REQUIRED	REQUIRED	MIL-S-48685
w.	Transistor, 2N6193	12630838	REQUIRED	REQUIRED	MIL-S-48685

NA = Not Applicable

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